Description

Image



Caption

Zips.

The material

POM was first marketed by DuPont in 1959 as Delrin. It is similar to nylon but is stiffer, and has better fatigue and water resistance - nylons, however, have better impact and abrasion resistance. It is rarely used without modifications: most often filled with glass fiber, flame retardant additives or blended with PTFE or PU. The last, POM/PU blend, has good toughness. POM is used where requirements for good moldability, fatigue resistance and stiffness justify its high price relative to mass polymers, like polyethylene, which are polymerized from cheaper raw materials using lower energy input.

Compositional summary

(CH2-O)n

General properties

Density		1.39e3	-	1.41e3	kg/m^3
Price	*	1.28	-	1.35	GBP/kg
Date first used		1956			

Mechanical properties

Young's modulus		2.6	-	3.2	GPa
Shear modulus	*	0.933	-	1.15	GPa
Bulk modulus	*	4.41	-	4.63	GPa
Poisson's ratio	*	0.386	-	0.401	
Yield strength (elastic limit)		57.2	-	71.7	MPa
Tensile strength	*	71.5	-	89.6	MPa
Compressive strength	*	105	-	116	MPa

Elongation		15	-	75	% strain
Hardness - Vickers	*	17	-	22	HV
Fatigue strength at 10^7 cycles		22	-	26	MPa
Fracture toughness		3.8	-	4.2	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.0125	-	0.0154	

Thermal properties

Melting point		160	-	175	°C
Glass temperature		-60.2	-	-50.2	°C
Maximum service temperature		82.9	-	96.9	°C
Minimum service temperature		-50.2	-	-40.2	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity		0.221	-	0.239	W/m.°C
Specific heat capacity		1.28e3	-	1.31e3	J/kg.°C
Thermal expansion coefficient		110	-	198	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity		3.3e20	-	3e21	µohm.cm
Dielectric constant (relative permittivity)		3.6	-	3.8	
Dissipation factor (dielectric loss tangent)		9.5e-4	-	0.00105	
Dielectric strength (dielectric breakdown)		18.9	-	20.5	MV/m

Optical properties

Transparency		Opaque			

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		1	-	2	
Moldability		4	-	5	
Machinability		3	-	4	
Weldability		4	-	5	

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Excellent			
Soils, alkaline (clay)		Excellent			

Wine	Excellent

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Unacceptable
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Limited use
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Limited use
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Limited use
Phosphoric acid (85%)	Unacceptable
Sulfuric acid (10%)	Unacceptable
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Limited use
Crude oil	Acceptable
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Limited use
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent
Acetone	Limited use

Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Unacceptable
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Unacceptable

Durability: built environments

Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Poor

Durability: flammability

Flammability	Highly flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 3.04	-	3.36	kg/kg
Embodied energy (virgin grade production)	* 81.8	-	90.2	MJ/kg
Water usage	* 240	-	265	l/kg

Material processing: energy

Polymer extrusion, embodied energy	* 8.11	-	8.98	MJ/kg
Polymer molding (injection, blow), embodied energy	* 22.4	-	25.6	MJ/kg
Machining, coarse, embodied energy	* 19	-	21.4	MJ/kg
Machining, fine, embodied energy	* 27.8	-	32	MJ/kg
Machining, grinding, embodied energy	* 36.7	-	42.6	MJ/kg

Material processing: climate change

Polymer extrusion, climate change (CO2-eq)	* 0.534	-	0.593	kg/kg
Polymer molding (injection, blow), climate change (CO2-eq)	* 1.57	-	1.81	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.24	-	1.4	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.82	-	2.1	kg/kg
Machining, grinding, climate change (CO2-eq)	* 2.4	-	2.79	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 1	-	1.11	kg/kg
Embodied energy, recycling	* 15.5	-	17.1	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 15.8	-	16.6	MJ/kg
Combustion CO2	* 1.45	-	1.52	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Acetal, like most thermoplastics, is an oil derivative, but this poses no immediate threat to its use.

Recycle mark



Supporting information

Design guidelines

POM is easy to mold by blow molding, injection molding or sheet molding, but shrinkage on cooling limits the minimum recommended wall thickness for injection molding to 0.1mm. As manufactured, POM is gray but it can be colored. It can be extruded to produce shapes of constant cross section such as fibers and pipes. The high crystallinity leads to increased shrinkage upon cooling. It must be processed in the temperature range 190-230 C and may require drying before forming because it is hygroscopic. Joining can be done using ultrasonic welding, but POM's low coefficient of friction requires welding methods that use high energy and long ultrasonic exposure; adhesive bonding is an alternative. POM is a good electrical insulator. Without coPolymerization or the addition of blocking groups, POM degrades easily.

Technical notes

The repeating unit of POM is - (CH2O)n and the resulting molecule is linear and highly crystalline. Consequently, POM is easily moldable, has good fatigue resistance and stiffness, and is water resistant. In its pure form, POM degrades easily by dePolymerization from the ends of the polymer chain by a process called 'unzipping'. The addition of 'blocking groups' at the ends of the polymer chains or coPolymerization with cyclic ethers such as ethylene oxide prevents unzipping and hence degradation.

Typical uses

POM is more expensive than commodity polymers such as PE, so is limited to high performance applications in which its natural lubricity is exploited. It is found in fuel-system, seat-belt components, steering columns, window-support brackets and handles, shower heads, ballcocks, faucet cartridges, and various fittings, quality toys, garden sprayers, stereo cassette parts, butane lighter bodies, zippers, telephone components, couplings, pump impellers, conveyor plates, gears, sprockets, springs, gears, cams, bushings, clips, lugs, door handles, window cranks, housings, seat-belt components, watch gears, conveyor links, aerosols, mechanical pen and pencil parts, milk pumps, coffee spigots, filter housings, food conveyors, cams, gears, TV tuner arms, automotive underhood components.

Tradenames

Acetron, Delrin, Fulton, Latan, Lupital, Plaslube, Tenac, Thermocomp, Ultraform

Links

ProcessUniverse

Producers

Reference

Values marked * are estimates.
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